

YAONAN JIN

jinyaonan@ust.hk

yj10027.github.io

CAREER HISTORY

Hong Kong University of Science and Technology Assistant Professor, Department of Computer Science and Engineering	2026.06 – present
Huawei (Shanghai) Researcher (Huawei's Top Minds Scheme) Taylor Lab (formerly Theoretical Computer Science Lab)	2023.05 – 2026.05

EDUCATION

Columbia University PhD in Computer Science, advised by Xi Chen and Rocco Servedio Thesis: <i>Bayesian Auction Design and Approximation</i>	2019.09 – 2023.04
Hong Kong University of Science and Technology MPhil in Industrial Engineering and Logistics Management, advised by Qi Qi Thesis: <i>Tight Approximation Ratio of Anonymous Pricing</i>	2017.09 – 2018.12
Shanghai Jiao Tong University BEng in Computer Science and Technology	2013.09 – 2017.06

RESEARCH INTERESTS

Theoretical Computer Science
Economics and Computation
Combinatorial Optimization
Sublinear Computation

INVITED ARTICLES

- [S1] Yaonan Jin and Pinyan Lu
Settling the Efficiency of the First-Price Auction [\[PDF\]](#)
ACM SIGecom Exchanges, 20(2): 69–74, 2022
- [S2] Yaonan Jin, Pinyan Lu, Qi Qi, Zhihao Tang, and Tao Xiao
Tight Revenue Gaps among Simple and Optimal Mechanisms [\[PDF\]](#)
ACM SIGecom Exchanges, 17(2): 54–61, 2019

JOURNAL PAPERS

In theoretical computer science, authors are usually listed alphabetically by surname.
Student coauthors I have mentored are underlined.

- [J1] Yaonan Jin and Pinyan Lu
Benchmark-Tight Approximation Ratio of Simple Mechanism for a Unit-Demand Buyer [\[arXiv\]](#)
To appear in SIAM Journal on Computing
- [J2] Yaonan Jin and Pinyan Lu
First Price Auction is $1 - 1/e^2$ Efficient [\[arXiv\]](#)
Journal of the ACM, 70(5): 36:1–36:86, 2023

- [J3] Jarosław Błasiok, Peter Ivanov, Yaonan Jin, Chin Ho Lee, Rocco Servedio, and Emanuele Viola
Fourier Growth of Structured $\mathbb{F}_2$ -Polynomials and Applications [\[arXiv\]](#) [\[ECCC\]](#)
(**Invited Paper**) To appear in Theory of Computing
- [J4] Yaonan Jin, Shunhua Jiang, Pinyan Lu, and Hengjie Zhang
Tight Revenue Gaps among Multiunit Mechanisms [\[arXiv\]](#)
SIAM Journal on Computing, 51(5): 1535–1579, 2022
- [J5] Nick Gravin, Yaonan Jin, Pinyan Lu, and Chenhao Zhang
Optimal Budget-Feasible Mechanisms for Additive Valuations [\[arXiv\]](#)
ACM Transactions on Economics and Computation, 8(4): 21:1–21:15, 2020
- [J6] Yaonan Jin, Pinyan Lu, Zhihao Tang, and Tao Xiao
Tight Revenue Gaps among Simple Mechanisms [\[arXiv\]](#)
SIAM Journal on Computing, 49(5): 927–958, 2020

CONFERENCE PAPERS

- [C1] Xihan Deng, Yaonan Jin, Wenqian Wang, and Yuhao Zhang
The Price of Anarchy for Selfish Load Balancing under Correlated Priors
International Joint Conference on Theoretical Computer Science–Frontier of Algorithmic Wisdom (IJTCS-FAW 2026)
- [C2] Xihan Deng, Yaonan Jin, Wenqian Wang, and Yuhao Zhang
The Price of Anarchy for Selfish Load Balancing
67th IEEE Symposium on Foundations of Computer Science (FOCS 2026)
- [C3] Sayan Bhattacharya, Martín Costa, Ermiya Farokhnejad, Shaofeng H.-C. Jiang, Yaonan Jin, and Jianing Lou
Fully Dynamic Euclidean k -Means [\[arXiv\]](#)
67th IEEE Symposium on Foundations of Computer Science (FOCS 2026)
- [C4] Yaonan Jin and Yingkai Li
Anonymous Pricing in Large Markets [\[arXiv\]](#)
27th ACM Conference on Economics and Computation (EC 2026)
- [C5] Houshuang Chen, Yaonan Jin, Pinyan Lu, and Chihao Zhang
Tight Regret Bounds for Fixed-Price Bilateral Trade [\[arXiv\]](#)
53rd International Colloquium on Automata, Languages, and Programming (ICALP 2026)
- [C6] Houshuang Chen, Yaonan Jin, Pinyan Lu, and Chihao Zhang
The Query Complexity of Uniform Pricing [\[arXiv\]](#)
35th ACM Web Conference (WWW 2026)
- [C7] Shaofeng H.-C. Jiang, Yaonan Jin, Jianing Lou, and Pinyan Lu
Local Search for Clustering in Almost-linear Time [\[arXiv\]](#)
37th ACM-SIAM Symposium on Discrete Algorithms (SODA 2026)
- [C8] Yiding Feng and Yaonan Jin
Beyond Regularity: Simple versus Optimal Mechanisms, Revisited [\[arXiv\]](#)
66th IEEE Symposium on Foundations of Computer Science (FOCS 2025)
- [C9] Yaonan Jin and Pinyan Lu
Benchmark-Tight Approximation Ratio of Simple Mechanism for a Unit-Demand Buyer [\[arXiv\]](#)
65th IEEE Symposium on Foundations of Computer Science (FOCS 2024)
- [C10] Xi Chen, Yaonan Jin, Tim Randolph, and Rocco Servedio
Subset Sum in Time $2^{n/2}/\text{poly}(n)$ [\[arXiv\]](#)

- [C11] Yaonan Jin, Pinyan Lu, and Tao Xiao
Learning Reserve Prices in Second Price Auctions [\[arXiv\]](#)
14th Innovations in Theoretical Computer Science Conference (ITCS 2023)
- [C12] Yaonan Jin, Daogao Liu, and Zhao Song
Super-Resolution and Robust Sparse Continuous Fourier Transform in Any Constant Dimension: Nearly Linear Time and Sample Complexity [\[arXiv\]](#)
34th ACM-SIAM Symposium on Discrete Algorithms (SODA 2023)
- [C13] Yaonan Jin and Pinyan Lu
The Price of Stability for First Price Auction [\[arXiv\]](#)
34th ACM-SIAM Symposium on Discrete Algorithms (SODA 2023)
- [C14] Yaonan Jin and Pinyan Lu
First Price Auction is $1 - 1/e^2$ Efficient [\[arXiv\]](#)
63rd IEEE Symposium on Foundations of Computer Science (FOCS 2022)
- [C15] Xi Chen, Yaonan Jin, Tim Randolph, and Rocco Servedio
Average-Case Subset Balancing Problems [\[arXiv\]](#)
33rd ACM-SIAM Symposium on Discrete Algorithms (SODA 2022)
- [C16] Jarosław Błasiok, Peter Ivanov, Yaonan Jin, Chin Ho Lee, Rocco Servedio, and Emanuele Viola
Fourier Growth of Structured $\mathbb{F}_2$ -Polynomials and Applications [\[arXiv\]](#) [\[ECCC\]](#)
25th International Workshop on Randomization and Computation (RANDOM 2021)
- [C17] Yaonan Jin, Shunhua Jiang, Pinyan Lu, and Hengjie Zhang
Tight Revenue Gaps among Multi-Unit Mechanisms [\[arXiv\]](#)
22nd ACM Conference on Economics and Computation (EC 2021)
- [C18] Yaonan Jin, Weian Li, and Qi Qi
On the Approximability of Simple Mechanisms for MHR Distributions [\[PDF\]](#)
15th Conference on Web and Internet Economics (WINE 2019)
- [C19] Nick Gravin, Yaonan Jin, Pinyan Lu, and Chenhao Zhang
Optimal Budget-Feasible Mechanisms for Additive Valuations [\[arXiv\]](#)
20th ACM Conference on Economics and Computation (EC 2019)
- [C20] Yaonan Jin, Pinyan Lu, Qi Qi, Zhihao Tang, and Tao Xiao
Tight Approximation Ratio of Anonymous Pricing [\[arXiv\]](#)
51st ACM Symposium on Theory of Computing (STOC 2019)
- [C21] Yaonan Jin, Pinyan Lu, Zhihao Tang, and Tao Xiao
Tight Revenue Gaps among Simple Mechanisms [\[arXiv\]](#)
30th ACM-SIAM Symposium on Discrete Algorithms (SODA 2019)

MANUSCRIPTS

- [M1] Yaonan Jin, Wei Tang, Yifan Wang, and Mengxiao Zhang
Query-Sample Tradeoffs for Revenue Maximization
- [M2] Shaofeng H.-C. Jiang, Yaonan Jin, Jianing Lou, and Weicheng Wang
Round-Efficient Fully-Scalable MPC Algorithms for k -Means [\[arXiv\]](#)
- [M3] Yaonan Jin
Tight Regret Bounds for Bilateral Trade under Semi Feedback [\[arXiv\]](#)
- [M4] Yaonan Jin, Yingkai Li, Yining Wang, and Yuan Zhou

RESEARCH FUNDING

- Huawei, Computing System Theory and Technology
Title: *Welfare and Revenue Guarantees of Practical Auctions*
Role: Project Lead, Period: 2024.01 – 2025.12, Amount: 3,526,000 CNY

PROFESSIONAL SERVICES

Program Committee

- WINE 2026, ICALP 2026, EC 2026, WWW 2026, WINE 2025, EC 2025

Journal Reviewer

- SIAM Journal on Computing, Algorithmica, Artificial Intelligence, ACM Transactions on Economics and Computation, Operations Research Letters

Conference Reviewer

- FOCS, STOC, SODA, EC, ITCS, ISAAC, WWW, WINE, SAGT

Service to Huawei's Taylor Lab

- Panelist: Huawei Collaborative Research Project (with University Faculty), 6 panels
- Theory seminar organizer, 2023.05 – 2026.05

SELECTED HONORS

Featured in Huawei 2024 Annual Report (p. 68) [PDF] for Research Highlights on [C9]	2025
Paper [C14] selected for “Highlights Beyond EC”, ACM EC	2023
Invited Survey [S1] in ACM SIGecom Exchanges for Research Highlights on [C14]	2022
Invited Survey [S2] in ACM SIGecom Exchanges for Research Highlights on [C20][C21]	2019
National Scholarship (1st Prize), Shanghai Jiao Tong University	2014

TEACHING EXPERIENCE

Columbia University

- COMS 4246: Introduction to Computational Complexity Fall 2020
Graduate course in complexity theory (TA for Xi Chen)
- COMS 4995: Incentives in Computer Science Spring 2020
Graduate course in algorithmic economics (TA for Tim Roughgarden)

Shanghai University of Finance and Economics

- 1665/101389: Discrete Mathematics Spring 2019
Undergraduate course in discrete mathematics (TA for Nick Gravin)
- 0008/213583: Mathematical Tools and Efficient Algorithms Summer 2018
Graduate course in graph and algebraic algorithms (TA for Richard Peng)

Hong Kong University of Science and Technology

- IEDA 3300: Industrial Data Systems Fall 2018
Undergraduate course in databases (TA for Qi Qi)

- IEDA 3300: Industrial Data Systems Spring 2018
Undergraduate course in databases (TA for Qi Qi)

Shanghai Jiao Tong University

- CS 358: Data Structures Fall 2016
Undergraduate course in data structures (TA for Yong Yu)
- CS 358: Data Structures Fall 2015
Undergraduate course in data structures (TA for Yong Yu)

TALKS GIVEN

- *Query-Sample Tradeoffs for Revenue Maximization*
 - Tsinghua University, Yau Mathematical Sciences Center, September 2026
- *The Price of Anarchy for Selfish Load Balancing*
 - Peking University, Center on Frontiers of Computing Studies, September 2026
 - CCF Annual Conference on Computational Economics, September 2026
- *Anonymous Pricing in Large Markets*
 - Institute of Software (CAS), Theory Seminar, September 2026
 - Renmin University of China, Gaoling School of Artificial Intelligence, September 2026
 - Great Bay University, Theory Seminar, July 2026
 - Shanghai Jiao Tong University, Theory Seminar, June 2026
- *Tight Regret Bounds for Fixed-Price Bilateral Trade*
 - National University of Singapore, Theory Seminar, January 2026
 - Tsinghua University, Yau Mathematical Sciences Center, November 2025
 - Renmin University of China, Gaoling School of Artificial Intelligence, November 2025
 - SUFE ITCS Workshop, June 2025
 - Huawei's Taylor Lab, Theory Seminar, May 2025
 - CUHK, Theory Seminar, April 2025
 - HKUST, Joint IEDA/CSE Seminar, April 2025
 - HKU, Theory Seminar, April 2025
- *Local Search for Clustering in Almost-linear Time*
 - CUHK-Shenzhen, Theory Seminar, July 2026
 - Nanyang Technological University, Theory Seminar, January 2026
 - New York University Shanghai, Theory Seminar, December 2025
 - Institute of Software (CAS), Theory Seminar, November 2025
 - **Invited Talk**, HCP 2025, August 2025
- *Beyond Regularity: Simple versus Optimal Mechanisms, Revisited*
 - FOCS 2025, December 2025

- Peking University, Center on Frontiers of Computing Studies, November 2025
- New York University Shanghai, Theory Seminar, October 2025
- City University of Hong Kong, Theory Seminar, April 2025
- Nanyang Technological University, Theory Seminar, November 2024
- National University of Singapore, Department of Economics, November 2024
- HKUST-Guangzhou, Information Hub, November 2024
- Shanghai Jiao Tong University, Theory Seminar, November 2024
- Huawei’s Taylor Lab, Theory Seminar, November 2024
- *Benchmark-Tight Approximation Ratio of Simple Mechanism for a Unit-Demand Buyer*
 - National University of Singapore, Theory Seminar, November 2024
 - FOCS 2024, October 2024
 - HKUST, IEDA, Theory Seminar, August 2024
 - **Invited Talk**, IJTCS 2024, July 2024
 - SUFE ITCS Workshop, June 2024
 - Renmin University of China, Gaoling School of Artificial Intelligence, December 2023
 - Institute of Computing Technology (CAS), Theory Seminar, December 2023
 - Tsinghua University, Yau Mathematical Sciences Center, December 2023
 - Peking University, Center on Frontiers of Computing Studies, December 2023
 - University of Science and Technology of China, Theory Seminar, December 2023
 - Huawei’s Taylor Lab, Theory Seminar, November 2023
- *Bayesian Auction Design and Approximation*
 - Columbia PhD Thesis Defense, April 2023
 - Columbia PhD Thesis Proposal, January 2023
- *The Price of Stability for First Price Auction*
 - SODA 2023, January 2023
- *Learning Reserve Prices in Second Price Auctions*
 - ITCS 2023, January 2023
- *First Price Auction is $1 - 1/e^2$ Efficient*
 - **Invited Tutorial**, COCOON 2024, August 2024
 - Tsinghua University, Institute for Interdisciplinary Information Sciences, December 2023
 - University of Science and Technology of China, Theory Seminar, December 2023
 - Shanghai Jiao Tong University, Theory Seminar, December 2023
 - Fudan University, School of Data Science, November 2023
 - **Invited Talk**, EC 2023 “Highlights Beyond EC” Sessions, June 2023
 - Institute of Computing Technology (CAS), Theory Seminar, December 2022

- TCS+ Talk, November 2022
- FOCS 2022, October 2022
- Harvard EconCS Seminar, October 2022
- UPenn Theory Seminar, October 2022
- Princeton Theory Seminar, October 2022
- Columbia Theory Seminar, September 2022
- *Bayesian Multi-Unit Revenue Maximization*
 - Columbia PhD Candidacy Exam, May 2022
- *Tight Revenue Gaps among Multi-Unit Mechanisms*
 - EC 2021, July 2021
- *On the Approximability of Simple Mechanisms for MHR Distributions*
 - WINE 2019, December 2019
- *Optimal Budget-Feasible Mechanisms for Additive Valuations*
 - EC 2019, June 2019
 - SUFE Theory Seminar, April 2019
- *Tight Approximation Ratio of Anonymous Pricing*
 - CCF TCS PhD Forum, June 2021
 - STOC 2019, June 2019
 - SUFE Theory Seminar, April 2019
 - HKUST, IEDA, Reading Group, December 2018
- *Tight Revenue Gaps among Simple Mechanisms*
 - Northwestern Theory Seminar, March 2019
 - SODA 2019, January 2019
 - HKU Theory Seminar, December 2018
 - SUFE Theory Seminar, December 2017
 - HKUST, IEDA, Reading Group, October 2017